

Activity 3: API Exploration and Integration using AI

Purpose

Learn how to use AI to understand REST APIs, interpret API documentation, generate integration code, and prepare an existing Oracle JET application for backend integration.

Pre-requisites

- Completion of Activities 1 and 2
- Working EMI Calculator application
- VS Code with Cline configured

Scenario

The SmartBank backend team has completed the **EMI Calculation REST API**. Your team has been asked to replace the client-side EMI calculation with a backend service in the next sprint.

Before implementing the integration, you need to understand the API, identify the required changes, and prepare the frontend for integration using AI.

Note: The objective of this activity is to understand and prepare API integration using AI. Implementing a live backend service is outside the scope of this activity.

Tasks

Task 1: Analyse the Existing Application

Understand how the current application performs EMI calculation.

Suggested Prompt

Analyze the current Oracle JET EMI Calculator.

Identify:

- Where the EMI calculation is currently performed.
- Which function contains the business logic.
- Which data should be sent to a backend API.
- Which values should be received from the API.
- What parts of the application would require modification for API integration.

Explain your findings before suggesting any code changes.

Task 2: Explore the REST API

Assume the backend team has shared the following API specification.

Sample API

POST /api/emi/calculate

Request

```
{
  "loanAmount": 500000,
  "annualInterestRate": 8.5,
  "loanTenure": 10
}
```

Response

```
{
  "monthlyEmi": 6199.64
}
```

Suggested Prompt

Analyze the following REST API specification.

Explain:

- Purpose of the API
- HTTP method
- Endpoint
- Request body
- Response body
- Expected HTTP status codes
- Possible error responses
- Input validation considerations

Present the explanation in simple language suitable for a beginner developer.

Task 3: Generate Integration Code

Generate the frontend code required to call the API.

Suggested Prompt

Using the current Oracle JET and KnockoutJS application, generate the code required to integrate the EMI Calculation REST API.

Requirements:

- Use the Fetch API with async/await.
- Send the loan amount, interest rate, and loan tenure to the backend.
- Read the monthly EMI from the response.
- Update the existing Knockout observable.
- Preserve the current UI.
- Include basic error handling.

Explain the changes before generating the updated code.

Task 4: Review the Integration

Review the generated integration before implementation.

Suggested Prompt

Review the proposed API integration.

Evaluate:

- Code quality
- Error handling
- Readability
- Maintainability
- Oracle JET best practices

Identify any improvements before the code is implemented.

Do not modify the generated code.